



Car Price Analytics

An end-to-end data analytics and machine learning solution for car valuation, exploratory analysis, and interactive dashboarding.



Data Analytics



Machine Learning



Streamlit Dashboard



Project Overview



End-to-End Pipeline

Complete lifecycle from raw data acquisition to model deployment and visualization.



Google Colab Engine

Data cleaning, feature engineering, statistical EDA, and machine learning model training.



Streamlit App

Interactive UI supporting feature filtering, vehicle comparison, and instant valuation.

Problem Statement

Complex Specification Dependencies

Vehicle prices depend non-linearly on horsepower, speed, torque, displacement, and fuel type.

Unstructured Raw Automobile Data

Inconsistent formatting, missing values, and mixed string ranges impact directly on decision making.

Lack of Automated Price Estimation

Traditional manual valuation lacks objective statistical predictive accuracy and interactive benchmarking.



Key Objectives

01

Clean & Preprocess Dataset

Format headers, handle null values, convert string ranges, and generate numeric columns.

02

Conduct Exploratory Analysis

Investigate relationships between fuel, speed, horsepower, manufacturer, and pricing.

03

Build Regression Models

Train Linear Regression and Random Forest models to accurately estimate vehicle valuation.

04

Deploy Streamlit UI

Build an interactive Web dashboard for car exploration, side comparison, and prediction.

Dataset & Features Breakdown

Vehicle Identity

- Car Name:** Specific model label
- Manufacturer:** Brand tag
- Fuel Type:** Petrol/Diesel/EV
- Engine Type:** Cylinder config

Performance Metrics

- Horsepower:** Engine power output
- Top Speed:** Max velocity (km/h)
- 0–100 km/h:** Acceleration time
- Torque:** Rotational force

Cleaned Features

- CC_Battery:** Displacement / EV size
- Seats_Clean:** Integer seating count
- Torque_Nm:** Standardized torque
- Target:** Cleaned Price (USD/INR)

Data Preprocessing Protocol

STEP 1

Header Sanitization

Removed extraneous spaces and standardizing casing across all column headers.

STEP 2

Numeric Formatting & Range Parsing

Converted string fields with ranges into single numerical representative values.

STEP 3

Missing Value & Duplicate Resolution

Imputed sparse missing values and purged duplicate vehicle records.

STEP 4

Feature Engineering

Engineered standardized columns: CC_Battery, Seats_Clean, and Torque_Nm.

Exploratory Data Analysis (EDA)



Fuel Type Impact

Analyzed average price variance between Petrol, Diesel, Hybrid, and EV models.



Horsepower Correlation

Strong positive correlation confirmed between engine power and total vehicle valuation.



Price Distribution

Identified right-skewed pricing distribution across luxury vs standard vehicle tiers.



Brand Positioning

Evaluated manufacturer car counts alongside brand-wise premium tier averages.



Machine Learning Architecture

BASELINE

Linear Regression

Evaluated as an initial linear benchmark for pricing trends based on numeric parameters.

✓ Easy interpretability

CHOSEN MODEL

Random Forest Regressor

Captures complex non-linear relationships and feature interactions across automotive specs.

✓ Superior prediction accuracy

DEPLOYMENT

Dashboard Serialization

Serialized via Joblib and loaded into Streamlit for sub-second user predictions.

✓ Seamless model persistence

End-to-End Project Workflow



Interactive Streamlit Dashboard

🔽 Dynamic Sidebar Filtering

Real-time search filters for manufacturer, fuel type, price slider, and horsepower thresholds.

📊 Executive KPI Scorecards

Displays total cars, mean pricing, average horsepower, and top speed metrics instantaneously.

📈 Interactive Visualizations

Plotly-powered interactive charts render smooth exploratory data drill-downs.



Dashboard Highlights & Sections



1. Overview

Fuel distribution, price distributions, and dataset-level aggregates.



2. Performance

Scatter plots for Horsepower, Top Speed, and Acceleration vs. Price.



3. Compare

Side-by-side spec comparison tool with dynamic radar overlay.



4. Manufacturers

Brand-wise market share, model count, and average price range.



5. Car Explorer

Inspect individual vehicle technical specs and torque charts.



6. Price Prediction

Input specifications to obtain instant Random Forest price predictions.

Real-Time Price Prediction Engine

- 1 Joblib Model Deserialization**
Streamlit initializes by loading the pre-trained Random Forest model state.

- 2 User Input Forms**
Users enter parameters: Manufacturer, HP, Speed, Torque, Fuel Type, and CC.

- 3 Instant Valuation Computation**
Model processes inputs and outputs an estimated vehicle market value instantly.



Application & Cloud Hosting



Development Layer

Google Colab notebook with Python, Pandas, Scikit-Learn, and Joblib serialization.



Source Control

GitHub repository for seamless application versioning and automatic Streamlit deployment triggers.



Cloud Host

Deployed on Streamlit Cloud for instant public browser accessibility.

Results & Key Takeaways



Visual Exploration

Car specifications can be easily explored via rich interactive Plotly chart widgets.



Pricing Pattern Insights

Identified key driver metrics across fuel categories, torque ratings, and manufacturer tiers.



Comparative Analysis

Users can instantly compare individual vehicles using multi-dimensional radar visualizations.



Unified Solution

Successfully integrated data engineering, ML modeling, and Web dashboarding in one solution.

Future Roadmap & Conclusion

Advanced Algorithms

Test XGBoost, LightGBM, and Neural Networks for lower prediction RMSE.

Recommendation Engine

Recommend similar car models based on budget and preferred specs.

Depreciation Analysis

Integrate vehicle age and mileage decay metrics for resale estimation.

THANK YOU!

<https://car-price-analytics-dashboard-dgx7vmvcjrkqohimxns2ap.streamlit.app/>